

QCML Research

I applied physics math to find hidden patterns in financial data

Problem

Can we detect market regime shifts *without labeled crisis data*?

Approach

Embed time series into geometric space using spectral metric learning

Extract 3 geometric observables measuring manifold deformation during stress

Fully unsupervised — no crisis labels, no look-ahead

Berry Phase z-score

QCML (Unsupervised)

Outcome

Multi-Lag Fidelity (**d=1.44**) beat Random Forest (**d=1.13**, p=0.002)

Tested across **5 ETFs** and **4 out-of-sample crises**

Solo-authored **34-page research paper** (in revision)

PythonPyTorchNumPy/SciPyOptunaLaTeX

Quanta Ventures

Developed systematic equity strategies for a quantitative fund

Problem

Building robust multi-strategy portfolios that survive regime changes

Approach

Modular framework: multiple strategy sleeves with volatility targeting

VIX/VVIX regime diagnostics for dynamic leverage

Walk-forward validation with 30-day embargo and robustness suite

2.92

Sharpe Ratio

5.02

Calmar Ratio

-0.070

Max Drawdown

133

Total Signals

Strategy Sleeves:

5 (Momentum, Mean Rev, Statistical, Risk Budget, Factor)

Validation:

Walk-forward with 30-day embargo + 11-test robustness suite

Outcome

Sharpe 2.92 and Calmar 5.02 on out-of-sample data

Passed full robustness test suite

Built C++ Heston pricer for options strategy component

Pythonscikit-learnOptunaPostgreSQLC++

Student Quant Fund

Finding alpha and building community

Problem

Giving underclassmen access to quantitative skills and a production trading platform

Approach

Teaching momentum and mean-variance strategies

Building production-ready platform for future cohorts

Preparing for paper trading and capital deployment

Engineering 7

Physics 9

ML

Outcome

Mentoring newer students in collaborative strategy development

Building institutional knowledge across cohorts

Platform designed for live paper trading

PythonJupyterTeam Collaboration

Skills

Hard Skills

Tech

Python (NumPy, Pandas, PyTorch)SQLGit/CI-CDockerLaTeXMathematica

CS

Data Structures & AlgorithmsML/PyTorchNumerical Optimization

Finance

Stochastic CalculusOptions Pricing (BS, Heston)Backtesting/Walk-ForwardRisk Metrics (Sharpe, VaR, CVaR)

Soft Skills

CS Teaching Assistant

Held weekly office hours, mentored 60+ students

Independent Researcher

Solo-authored 34-page research paper

Team Collaboration

Student Quant Fund leadership

Technical Communication

Translating complex math for diverse audiences

Pitzer College | Keck Science Department

Carpe Diem Endowed Scholar

Seeking: Quant Research/Trading | ML/AI Engineering | Physics Research

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